Blood Pressure Variability: Its Measurement and Significance in Hypertension

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Blood pressure (BP) fluctuations over time physiologically result from the complex interaction between environmental stimulation, genetic factors, and cardiovascular control mechanisms. Ambulatory BP-monitoring techniques, in particular systems providing beat-by-beat BP recording, have allowed a detailed description of the different components contributing to overall BP variability (BPV) over 24 hours, including short-lasting and more sustained BP changes. In hypertension, BPV increases with increasing BP levels, and evidence is available that its magnitude correlates closely with target-organ damage and with the incidence of cardiovascular events, independent of absolute BP levels. It has been suggested that drugs capable of providing smooth 24-hour BP control, reducing BPV, may confer additional target-organ protection. Mathematic indices, such as the trough-to-peak ratio and the smoothness index, represent useful measures of the homogeneity of the antihypertensive effect over 24 hours. Further studies are still needed to confirm that, in humans, interventions that can reduce BPV can also reduce the rate of cardiovascular events.

Introduction

Evidence collected during the past 30 to 40 years, in both pathophysiologic and clinical studies, has clearly shown that useful information may arise from analysis of blood pressure variability (BPV). In this paper, we briefly summarize such evidence and facilitate an understanding of the clinical relevance of assessing this parameter in subjects with hypertension. In particular, the possible role of BPV, both as an independent contributor to hypertension morbidity and mortality and as a target of antihypertensive treatment, is specifically addressed.

Assessment of Blood Pressure Variability

Physiologically, BP undergoes continuous fluctuations over time as a result of a complex interaction among cardiovascular control mechanisms. The occurrence of these fluctuations was first described at the beginning of the 18th century by Stephen Hales (1733), but the assessment of BPV in the clinical setting was made possible only by the end of the 19th century, thanks to the sphygmomanometric technique introduced by the Italian scientist Scipione Riva Rocci [1]. In the 1960s, the development of the intra-arterial Oxford System, allowing a continuous ambulatory BP monitoring [2], represented a further step forward in BPV assessment in humans. In fact, by making the measurement of beat-to-beat BP changes in freely moving subjects possible, it offered a way to accurately analyze the BP variations occurring over the whole 24-hour period [3•]. It was by using this invasive technique that different components of BPV could be identified. They range from short-lasting changes to long-lasting circadian fluctuations, the latter being characterized by a decrease in BP during the night-time followed by a sharp increase in the morning, and by another decrease over the siesta time.

More recently, noninvasive ambulatory BP monitoring (ABPM) techniques, based on automated oscillometric arm-cuff BP readings, have become increasingly available [4]. As these methods yield BP values every 15 to 30 minutes, they are unable to precisely quantify short-lasting BP changes, for which BP readings at intervals equal or shorter than 10 minutes would be required [5]. In spite of its relative inaccuracy, the noninvasive assessment of BPV by conventional ABPM has been widely used in clinical practice. It has allowed a better assessment on a far larger scale of both the BP behavior over the 24-hour period and the BPV role as a predictor of hypertension consequences, such as target-organ damage and cardiovascular events. However, the discontinuous nature of BP sampling has prevented a detailed assessment of BPV phenomena. This limitation has been recently overcome by the introduction of a noninvasive technique for continuous 24-hour ABPM at the finger level (Portapres device; Finapres Medical Systems, Amsterdam, The Netherlands) [6].

Over the years, several methods have been proposed to analyze the ABP tracings provided by the invasive and noninvasive techniques discussed earlier, aimed at providing a quantitative description of the BP changes

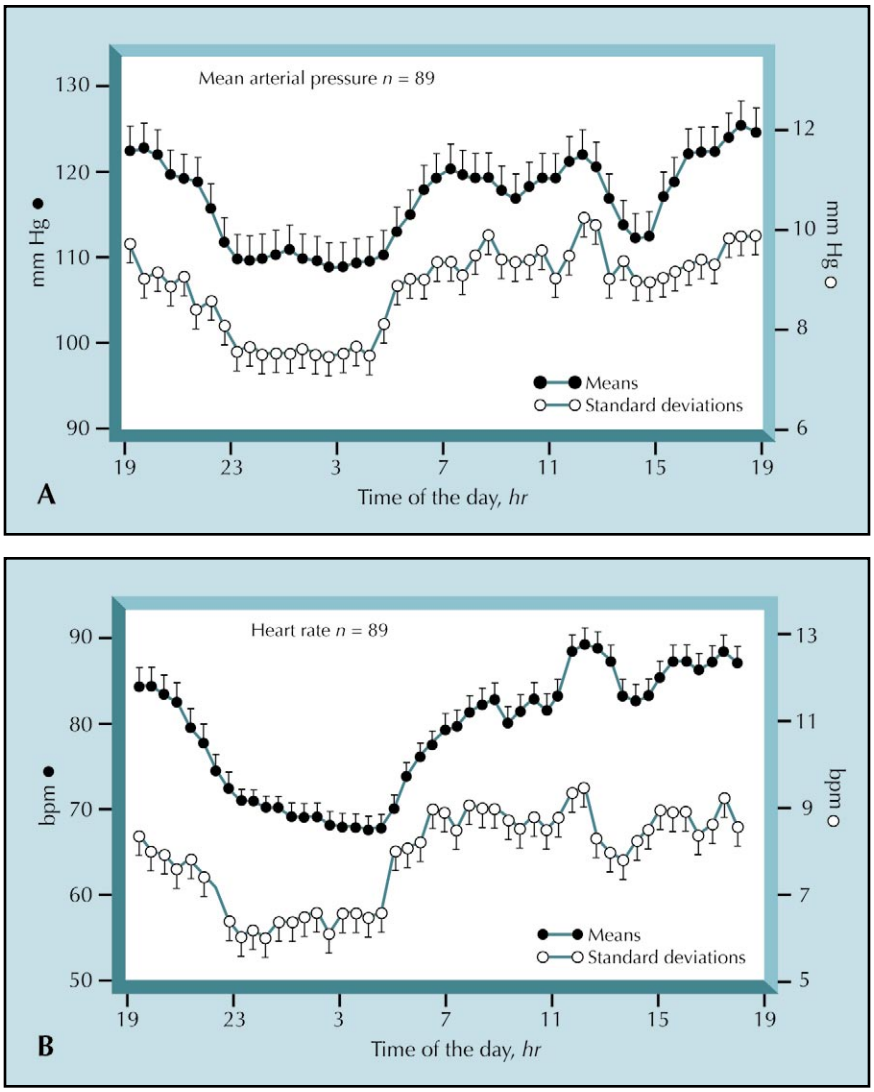

Figure 1. Blood pressure and heart rate variability over 24 hours in a group of 89 hypertensive patients. *Black circles* refer to average half-hour values and *white circles* refer to half-hour standard deviations. Data are separately shown for mean blood pressure (**A**) and heart rate (**B**). (*Adapted from* Mancia et al. [3•].)

occurring over the day and night. These methods include the straightforward computation of the BP variations induced by specific behaviors or associated with shifts between selected subperiods within the 24 hours (ie, between day and nighttime or in the early morning). They also include more complex calculations, such as those based on spectral analysis, focusing on BP fluctuations in the frequency domain [7,8]. Among these different approaches, in both pathophysiologic and clinical studies, the method most commonly employed for BPV assessment has been, and still is, calculation of the standard deviation (SD) of average BP values over the 24-hour, the daytime, and the nighttime BP periods.

When considering the analysis of continuous BP recordings, further development of the approach based on calculation of BP SD has allowed separate quantification of short- and long-lasting BPV components. Assessment of long-term BP changes is provided by calculating mean BP values, with the related SDs, for each of the 48 half hours into which a 24-hour BP recording can be subdivided. When the 48 half-hour mean BP values are averaged, the resulting SD (among half-hour SDs, mostly reflecting circadian variations) is taken as a measure of long-term BPV. Short-term BPV is conversely quantified as the average of the 48 SDs of each 30-minute segment (within half-hour SD). Based on these observations, it was shown that BPV is greater during the daytime and lower during the nighttime or during resting hours (Fig. 1).

It was also shown that BPV is progressively greater with the increase in BP mean levels as well as with advancing age, at least for short-term BP changes [3•].

Blood Pressure Variability in Hypertension

An increased BPV is a well-established feature of hypertension, both in experimental animals and in humans [8], and there is a growing body of evidence suggesting that the entity of BPV correlates to some extent with the severity of BP elevation. In our laboratory [3•], BPV was continuously assessed for 24 hours under standardized ambulatory conditions by the Oxford technique [9] in subjects classified as normotensive, mild hypertensive,

and severe hypertensive, according to their mean 24-hour BP values. Both the short-term (within half-hour SD) and the long-term (among half-hour SD) BPV values were greater in hypertensives compared to normotensives. Additionally, in both groups, also within each subject, the amplitude of BP fluctuations changed in parallel with the levels of mean BP achieved at different times of the 24 hours, confirming the association occurring between degree of BPV and BP levels.

Association of Blood Pressure Variability with Target-organ Damage and Cardiovascular Morbidity and Mortality

The possible independent contribution of an enhancement of BP fluctuations in determining the cardiovascular consequences of hypertension, in addition to the effect of an increase in mean BP levels, has been addressed by several studies. In most cases, although not in every instance, evidence has been provided that an increased BPV per se—that is, independent from an increase in mean BP levels—is a determinant of target-organ damage. This has been demonstrated both in the general population and in hypertensive patients. It was also shown when considering the degree of BPV occurring during the daytime [10••,11], the nighttime [12], or the overall 24-hour period [13•].

Data favoring an independent role of BPV in determining hypertensive organ damage first came from a study carried out in 108 mild-to-severe essential hypertensives by means of 24-hour intra-arterial ABPM. Both 24-hour mean BP and 24-hour BPV appeared to be related to prevalence and severity of target-organ damage, assessed as an overall score derived from patients' clinical evaluations [13•]. Overall, prevalence and severity of organ damage were more closely related to 24-hour mean BP and 24-hour BPV than to clinic BP. Interestingly, at any given 24-hour mean BP level, prevalence, and severity of target-organ damage were greater in patients with higher short-term and long-term BPV. These cross-sectional data were confirmed by longitudinal observations carried out in the same group of patients [14••], which provided the first solid evidence in favor of a predictive role played by BP variability in hypertension. In this study, 73 patients with essential hypertension had their ABP and BPV assessed intra-arterially for 24 hours with the Oxford technique and then were followed-up for 1 to 13 (mean 7.4) years. They were subdivided into four groups based on their average mean arterial pressure (MAP). Subjects within each group were further subdivided into two classes according to whether their long-term BPV (among half-hour SD) was lower or higher than the mean long-term BPV of the respective group. For each group, the two subclasses displayed similar mean BP levels, but left ventricular hypertrophy (LVH) was more prevalent in the subclass with higher BPV. Thus, entry long-term BPV was demonstrated to be a determinant of cardiovascular complications in hypertension (besides follow-up clinic BP and entry level of organ damage) even after accounting for differences in mean BP levels. These observations were confirmed by data obtained from experimental animals, whose BP oscillations were artificially increased by sinoaortic denervation (SAD)—that is, by surgical interruption of the carotid and aortic baroreceptor afferent fibers. A few weeks after SAD, experimental animals displayed an increase in BPV without a parallel increase in mean BP levels. Such an increased BPV was associated with both vascular and cardiac structural changes, even in presence of persisting normal BP levels [15–19].

The evidence favoring BPV as a determinant of target-organ damage was further strengthened by larger scale studies in humans, in which BPV was assessed non-invasively on sizable populations by means of 24-hour conventional ABPM. In more than 700 subjects, who were either normotensive or with essential hypertension of various severity, higher daytime systolic BP (SBP) and higher daytime SBP variability were related to higher degrees of vascular changes (on fundoscopic examination) and of LVH (on electrocardiogram and chest radiograph) [12].

In the European Lacidipine Study on Atherosclerosis (ELSA) study [20], 1663 hypertensives underwent 24-hour ABPM and were subsequently divided into quintiles according to their 24-hour mean BP. Each quintile was further divided into two subgroups, depending on whether BPV was greater or lower than the subgroup mean. Carotid artery intima-media thickness (IMT) assessed on duplicate ultrasound examinations was positively related to 24-hour, day and night mean SBP and diastolic BP (DBP), although the real novel observation of this study was in the demonstration of a strong and linear relationship between IMT and measures of overall 24-hour SBP and pulse pressure (PP) variability. The results of this study have relevant implications from the clinical standpoint, as they further support the clinical relevance of BPV, by showing that in subjects with hypertension, pathologic changes in large arteries correlate not only with mean 24-hour BP, but also with 24-hour BP variability.

Additionally, the PAMELA study [21] (Pressioni Arteriose Monitorate E Loro Associazioni) allowed further consolidation of the evidence in favor of an association between BPV and target-organ damage in 1648 untreated subjects who were randomly selected from the general population of a small city near Milan, Italy. In this sample, BPV was estimated by an oscillometric device every 20 minutes for 24 hours, and left ventricular mass index (LVMI) was assessed by echocardiography. From all of the systolic, diastolic, and heart-rate readings, a number of parameters were calculated: the 24-hour mean SBP and DBP; the SD of these 24-hour means (overall variability); the cyclic components of their variations, which when assessed by Fournier analysis explained more than 95% of the overall variability; and the fraction of the overall

SBP and DBP variability that was not accounted for by the main two cyclic components (referred to as "residual variability"). LVMI was significantly associated with both 24-hour mean SBP and DBP and with their residual BPV, thus confirming the clinical relevance of BPV in the general population, also.

The noninvasive 24-hour ABPM, carried out in large populations, allowed confirmation of the putative role of BPV in predicting not only organ damage occurrence and progression but also cardiovascular events. Sander et al. [10••] addressed this issue in a series of 286 hypertensives followed for 3.3 years whose circadian BP changes and early carotid atherosclerosis were assessed, respectively, by 24-hour ABPM and ultrasound measurement of IMT of the common carotid artery. Subjects with increased SBP variability had faster progression of IMT, even after adjustment for other risk factors. Furthermore, on multivariate analysis, daytime SBP variability appeared to be the best predictor of IMT progression and was associated with an increased risk for cardiovascular events.

Finally, in the Osahama study [11], both increased SBP variability and decreased heart-rate variability, as assessed by 24-hour ABPM in 1542 subjects from this Japanese population, appeared to be independent predictors of cardiovascular mortality, thus confirming in the general population what had already been seen in hypertensive patients [22,23].

Role of Blood Pressure Variability in Treatment of Hypertension

The independent association of increased BPV with both target-organ damage and cardiovascular complications in hypertension has suggested that BPV could become an additional target of antihypertensive treatment besides mean BP reduction per se. Indeed, some experimental evidence has been provided to support this concept. In a recent study [24], spontaneously hypertensive rats were treated with either nitrendipine (reducing both mean BP and BPV) or hydralazine (reducing only mean BP to the same extent of nitrendipine, but with no effect on BPV). The regression in target-organ damage was significantly more evident with nitrendipine than with hydralazine—that is, at comparable degrees of mean BP reduction, it was more pronounced with the drug able to reduce BPV, also. Additionally, in the group treated with nitrendipine, organ damage correlated significantly with BPV more than with mean BP level.

This implies that an ideal antihypertensive drug should reduce not only mean BP levels, but also BPV. A reduction in BPV could be achieved by buffering short-lasting BP fluctuations, a task sometimes difficult to achieve due to the important involvement of autonomic influences in the genesis of this phenomenon [7]. A reduction in overall BPV over 24 hours, however, could result also from use of antihypertensive drugs characterized by a smooth distribution of their antihypertensive effects throughout the dosing interval, and thus able to downshift an elevated 24-hour BP profile, preserving its physiologic circadian pattern. This is likely to be obtained when using long-acting drugs capable of smoothly exerting their action over 24 hours without any imbalance between peak and trough effects. Conversely, this is unlikely to be observed with the administration of short-acting compounds, which typically loose their BP-lowering effect before the end of the dosing interval—that is, at trough time. When short-acting drugs are scheduled in the morning, loss of their antihypertensive coverage may affect a crucial period of the 24 hours (ie, early the next morning). This may have clinical relevance because at this time, probably because of the sympathetically mediated increase of both BP and HR, patients experience the highest incidence of cardiovascular and cerebrovascular events [25•,26–28].

Such a possibility is exemplified by the results of a recent study in which administration of a short-acting compound scheduled in the morning was responsible for inadequate BP control at the end of the dosing period (next morning) in as many as 62% to 85% of patients [29].

Long-acting antihypertensive drugs appear to score better in this context, as shown by data collected with administration of drugs characterized by a prolonged half life, such as telmisartan, an angiotensin II–receptor blocker, or amlodipine, a dihydropyridine calcium channel blocker [30,31].

From these considerations, it appears that not only the mean amplitude of the BP reduction, but also the duration and the smoothness of the BP-lowering effect, are crucial aspects to pay attention to when evaluating the performance of an antihypertensive medication. To provide an assessment of these features, a few quantitative indices have been developed.

Trough:peak ratio

The trough-to-peak (T:P) ratio has been developed not only to measure the duration of the BP-lowering effect of an antihypertensive medication but also to assess the homogeneity of such an effect over the dosing interval [32]. Originally, it was proposed to be used with standardized sphygmomanometric BP measurements at rest. More commonly, however, it is derived from automated BP measurements taken out of a 24-hour ABPM. In the latter case, the T:P ratio is calculated by dividing the BP reduction at the end of the dosing interval (trough) by the BP reduction at the time of the maximum effect (peak), usually 2 to 8 hours after dosing. Although the ideal drug providing a homogeneous BP reduction throughout the dosing period should have a T:P ratio equal to 1, the United States Food and Drug Administration [33] pragmatically approves antihypertensive drugs with a T:P ratio down to 0.5.

Despite its usefulness, the T:P ratio is not free from limitations [8]. First, it focuses on very short time windows,

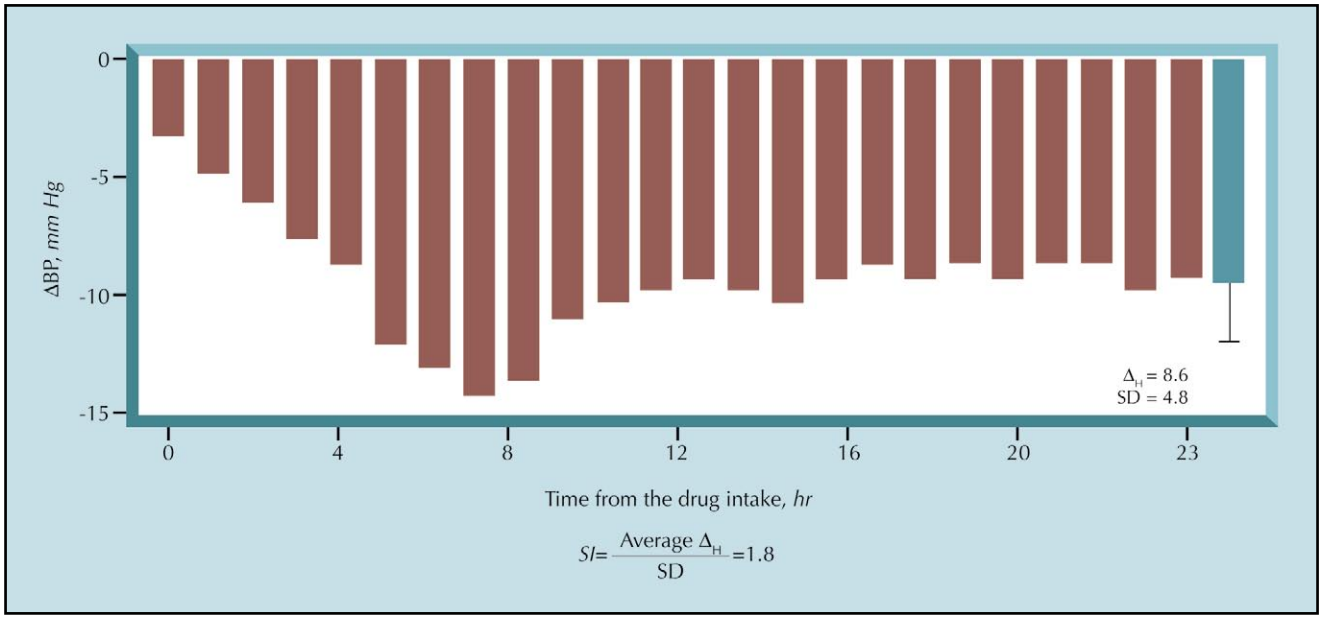

Figure 2. An example of the calculation of the smoothness index. Δ_H—average of the 24-hourly blood pressure changes between baseline and treatment; SD—standard deviation of the average of the 24-hourly blood pressure changes; SI—smoothness index. (*Adapted from* Parati et al. [34•].)

largely underutilizing the information provided by an ABPM and BP during the entire 24-hour period. Second, it may be influenced by patients' behaviors and posture and by spontaneous BP variations more than by the BP-lowering effect of a drug. Finally, it can be close to 1, even in the absence of any appreciable BP-lowering effect, whenever such a lack of BP reduction occurs, both at peak and at trough as, for example, in the case of placebo.

Smoothness index

The smoothness index (SI) has been developed in our laboratory [31] to overcome some of the limitations of the T:P ratio. From analysis of duplicated 24-hour ABPM (performed before and during pharmacologic treatment), the mean 24-hourly changes in BP are first calculated. The average of these 24-hourly changes, together with its SD, is then assessed. The SI is the ratio between the average of the 24-hourly BP changes induced by a given medication and its SD (Fig. 2). Evidence is available that use of this index provides a measure of both the amplitude and the distribution of the BP reduction obtained by a given drug that relates to drug-induced regression of target-organ damage better than the trough-peak ratio [34•].

Conclusions

An increased BPV appears to importantly contribute to target-organ damage and to the enhanced cardiovascular risk of hypertensive patients, over and above the effect of an increase in mean BP levels. Reducing 24-hour mean BP is the main aim of antihypertensive therapy, but data are available that additional cardiovascular protection can be achieved by reducing 24-hour BPV. However, to definitively prove the prognostic role of BPV and the need for its control by treatment, evidence is still needed from intervention trials aimed at demonstrating that by reducing BPV through administration of antihypertensive drugs, a reduction in organ damage and/or in the rate of cardiovascular events can be obtained. Such a demonstration, now available from animal studies only, might also provide clinicians with important practical indications on the choice of the most appropriate antihypertensive drug or drug combination to be used in cases of enhanced BP fluctuations.

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